Incidence operators, orthogonal codes and permutation modules

Common theory for two applications

The starting point is a symmetric operator on a space with its coordinate bilinear form. An orthogonal decomposition of its image can often be read from a short polynomial identity, or transferred from a much smaller incidence space. This separates the questions of dimension and duality from the questions of minimum weight and group action. The latter require additional arguments: averaging over coordinate permutations reduces a distance search, a matrix algebra certifies irreducibility, and refinement of a support incidence structure bounds the full automorphism group. Each argument is proved once here and then applied to literal matrices and finite data in the accompanying papers.

Our two source manuscripts are Leemans and Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Math. Commun. 21 (2016), 239–249 (L050), and Key and Rodrigues, *LCD codes from adjacency matrices of graphs*, the seventeen-page manuscript dated 4 August 2017 (F017). The graph constructions, original code assertions, and results quoted from their predecessors remain attributed to those sources in the applications. Orthogonal projections, incidence counting, the simplex chain contraction, MacWilliams duality and elementary module theory are classical tools. Our contribution here is their common derivation and the explicitly checked passage to the stated applications. Historical priority for these methods is not asserted.

The common complement theorem includes the possible one-dimensional defect and does not require an integral spectrum. The subset and Hamming calculations give families of consequences of the same operator tests. The binary incidence argument also determines all minimum words, whereas the ternary triple-incidence calculation corrects a dimension in F017. Other corrections, retained classification results and finite computation premises are stated with the exact source assertions in each application. No numerical compression factor is assigned by this draft.

The whole of this common paper is a prerequisite for the applications when lengths are compared. A theorem used by only one application is still charged here in full. Executable certificates have a separate inventory; any manuscript comparison excludes executable programs on both sides. The nine generic Lean declarations already associated with the projection and averaging arguments retain their limited stated scope. They do not formalize all the proofs in this paper.

1 Symmetric operators and graph duality

Let F be a field and let $V = F^n$, where $n > 0$. We use the standard nondegenerate symmetric bilinear form $\langle x, y\rangle = x^{\mathsf{T}}y$. For a symmetric matrix A, write $C_A = \operatorname{im} A$. Thus $C_A^{\perp} = \ker A$. The code C_A is LCD if $C_A \cap C_A^{\perp} = 0$. Write $j = (1, \dots, 1)^{\mathsf{T}}$ and $J = jj^{\mathsf{T}}$; integers in matrix identities denote their images in F.

Lemma 1.1. *If E is symmetric and $E^2 = E$, then $V = \operatorname{im} E \perp \oplus \ker E$. In particular $\operatorname{im} E$ is LCD. If $E_1, \dots, E_t$ are symmetric idempotents with $E_iE_k = 0$ for $i \neq k$ and $\sum E_i = I$, their images form an orthogonal direct sum equal to V.*

Proof. Every v is $Ev + (v - Ev)$, with the two terms in the asserted spaces. If $x = Ey$ and $Ex = 0$, then $x = E^2y = Ex = 0$. Moreover $\langle Ey, z\rangle = \langle y, Ez\rangle = 0$ when $Ez = 0$. For several idempotents, $v = \sum E_i v$ gives spanning, and application of E_k to a relation $\sum x_i = 0$, $x_i \in \operatorname{im} E_i$, gives $x_k = 0$. The same inner-product identity proves pairwise orthogonality. □

Theorem 1.2. *Let A be symmetric and put $B = J - I - A$. Suppose that $AB = 0$. There is a scalar $\nu \in F$ for which $Aj = \nu j$. Put $\bar{\nu} = n - 1 - \nu$. Then $\nu\bar{\nu} = 0$, both C_A and C_B are LCD, and the following are orthogonal decompositions with the indicated projection matrices:*

condition	*decomposition*	*projections*
$\nu \neq 0$	$V = C_A \perp\!\!\oplus C_B$	$J - A,\ -B$
$\bar{\nu} \neq 0$	$V = C_A \perp\!\!\oplus C_B$	$-A,\ J - B$
$\nu = \bar{\nu} = 0$	$V = C_A \perp\!\!\oplus C_B \perp\!\!\oplus Fj$	$-A,\ -B,\ J.$

Consequently $C_A = C_B^\perp$ if and only if $(\nu, \bar{\nu}) \neq (0, 0)$. In the exceptional case $C_B^\perp = C_A \perp\!\!\oplus Fj$ and $C_A^\perp = C_B \perp\!\!\oplus Fj$.

Proof. Transposition gives $BA = 0$. Expanding these two products gives

$$AJ = A^2 + A = JA.$$

If $d = Aj$, then $AJ = dj^{\mathsf{T}}$ and $JA = jd^{\mathsf{T}}$. Equality of their (r, s) entries says $d_r = d_s$ for all r, s. Since $n > 0$, this defines ν, and hence $Bj = \bar{\nu} j$. Applying $AB = 0$ to j gives $\nu\bar{\nu} = 0$. We have also proved

$$A^2 + A = \nu J, \quad B^2 + B = \bar{\nu} J, \quad AJ = JA = \nu J, \quad BJ = JB = \bar{\nu} J, \quad J^2 = nJ.$$

Suppose first that $\nu \neq 0$. Then $\bar{\nu} = 0$ and $n - \nu = 1$. Set $E = J - A$ and $F = -B$. Direct multiplication gives

$$E^2 = (n - \nu)J - A = E, \quad F^2 = B^2 = -B = F, \quad EF = FE = 0, \quad E + F = I.$$

Here $\operatorname{im} F = \operatorname{im} B$. Since $Aj = \nu j$ with $\nu \neq 0$, we have $j \in \operatorname{im} A$, so $\operatorname{im}(J - A) \subseteq \operatorname{im} A$. Conversely $EA = JA - A^2 = A$, so $\operatorname{im} A \subseteq \operatorname{im} E$. Lemma 1.1 proves the first row. Interchanging A and B proves the second.

Finally, if $\nu = \bar{\nu} = 0$, then $n = 1$ in F. The displayed identities show that $-A, -B, J$ are symmetric, pairwise annihilating idempotents, with sum I. Their images are C_A, C_B, Fj. Since $j \neq 0$ and $\langle j, j\rangle = n = 1$, the last summand is a nonzero nondegenerate line. This proves the third row and both descriptions of the orthogonal complements. In particular that line prevents equality $C_A = C_B^\perp$ in precisely this case. □

Corollary 1.3. *Let A be the adjacency matrix of a nonempty finite simple graph over F and B that of its complement. If the two graph codes are orthogonal, then both are LCD. The vertices have equal degrees in F, even when their integer degrees are not equal. The complement code is the full orthogonal dual unless both degrees vanish in F, in which case the missing summand is exactly the constant line.*

Proof. Orthogonality of the column spaces is $A^{\mathsf{T}}B = AB = 0$. Apply Theorem 1.2. Matrix multiplication by j records the vertex degrees. □

Proposition 1.4. *If A is symmetric and $A(A+I) = 0$, then $-A$ and $A+I$ are the orthogonal projections onto C_A and C_{A+I}, respectively. Thus $C_A = C_{A+I}^\perp$. Conversely this equality implies $A(A + I) = 0$.*

Proof. The two matrices are symmetric, sum to I, annihilate one another, and are idempotent because $A^2 = -A$. Their images are as stated, so Lemma 1.1 applies. For the converse, the asserted equality implies orthogonality of all columns, which gives $A(A + I) = 0$. □

Proposition 1.5. *Let A be the adjacency matrix of a noncomplete, nonempty strongly regular graph with parameters (v, k, λ, μ), having both an edge and a nonedge. Over a field of characteristic p, it satisfies*

$$A(A + I) = 0 \quad \Longleftrightarrow \quad k = 0, \quad \lambda = -1, \quad \mu = 0 \pmod{p},$$

and

$$A(J - I - A) = 0 \quad \Longleftrightarrow \quad k - \mu = 0, \quad k - 1 - \lambda = 0 \pmod{p}.$$

The first condition gives the reflexive-code decomposition of Proposition 1.4. The second gives the exact decomposition of Theorem 1.2, with $\nu = k$ and $\bar{\nu} = v - 1 - k$.

Proof. Counting common neighbours gives $A^2 = (k-\mu)I + (\lambda-\mu)A + \mu J$. Its diagonal, edge and nonedge entries give the first equivalence directly. For the second, the edge entries of $A(J - I - A)$ equal $k - 1 - \lambda$, the nonedge entries equal $k - \mu$, and the diagonal entries are zero. All three types occur under the stated assumptions. □

2 Incidence transfer and residual designs

Lemma 2.1. *Let H be an $m \times n$ matrix over a field, with nonsingular Gram matrix $G = HH^{\mathsf{T}}$. Let A be symmetric and suppose*

$$A = H^{\mathsf{T}} M H$$

for an $m \times m$ matrix M. Then

$$\operatorname{im} A = H^{\mathsf{T}}(\operatorname{im} M), \qquad \operatorname{rank} A = \operatorname{rank} M.$$

Under this identification the form on $\operatorname{im} A$ is the restriction to $\operatorname{im} M$ of $(x, y) \mapsto x^{\mathsf{T}} G y$. Consequently $\operatorname{im} A$ is LCD exactly when this restricted form is nondegenerate.

Proof. Nonsingularity of G implies that H is onto and H^{T} is injective: $H(H^{\mathsf{T}} G^{-1} x) = x$, and $H^{\mathsf{T}} x = 0$ implies $Gx = 0$. Hence MH has image $\operatorname{im} M$, and applying the injective map H^{T} proves the first two assertions. For $x, y \in \operatorname{im} M$,

$$\langle H^{\mathsf{T}} x, H^{\mathsf{T}} y \rangle = x^{\mathsf{T}} G y.$$

The radical of the restricted form on a code is its intersection with its orthogonal complement. This proves the last assertion. □

Let $\mathcal{D}$ be a Steiner system $S(4, 7, 23)$. Fix a set T of size $t \in \{0, 1, 2\}$. Retain the heptads disjoint from T, and write H_t for their point–block incidence matrix on the remaining $23 - t$ points. Two distinct heptads of $\mathcal{D}$ meet in either one or three points. This intersection property is a retained property of this Steiner system; the explicit binary construction used in our finite certificate also verifies it for every pair. Define A_t on the retained blocks by joining two distinct blocks precisely when they meet in three points. All matrix calculations in the next theorem take place over $\mathbb{F}_3$. Write e for the constant vector on points and j for the constant vector on blocks; the subscripts on identity and all-one matrices indicate their sizes.

Theorem 2.2. *The three graph codes have the following descriptions, dimensions and orthogonal decompositions:*

t	n	$H_tH_t^{\mathsf{T}}$	$\operatorname{im} A_t$	$\operatorname{rank} A_t$
0	253	$2I_{23}$	$\operatorname{im} H_0^{\mathsf{T}}$	23
1	176	$I_{22}+J_{22}$	$H_1^{\mathsf{T}}(e^{\perp})$	21
2	120	I_{21}	$\operatorname{im} H_2^{\mathsf{T}}$	21.

Each of these codes is LCD. Moreover, if $B_t = J_n - I_n - A_t$ is the complementary adjacency matrix, then

$$(\operatorname{im} A_t)^{\perp} = \operatorname{im} B_t \quad (t = 1, 2).$$

For $t = 0$, the code has the orthogonal splitting

$$\operatorname{im} A_0 = \mathbb{F}_3 j \perp \oplus H_0^{\mathsf{T}}(e^{\perp}),$$

whose second summand has dimension 22 and is LCD. The orthogonal dual of $\operatorname{im} A_t$ has dimension $n - \operatorname{rank} A_t$ in each case.

Proof. The numbers of heptads through an s-set, for $0 \le s \le 4$, are

$$\lambda_s = \frac{\binom{23-s}{4-s}}{\binom{7-s}{4-s}} \quad \text{and hence} \quad (\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (253, 77, 21, 5, 1).$$

Indeed, count the four-subsets containing that s-set and use the unique heptad containing each four-subset. Inclusion–exclusion on the points of T now gives the number n_t of retained blocks, the number r_t through each remaining point and the number s_t through each remaining pair:

t	n_t	r_t	s_t
0	253	77	21
1	176	56	16
2	120	40	12.

Thus $H_tH_t^{\mathsf{T}} = (r_t - s_t)I + s_tJ$, giving the three Gram matrices in the statement. They are nonsingular. Only the middle one needs comment: if $(I+J)x = 0$ and $u = e^{\mathsf{T}}x$, then $x = -ue$ and $u = -22u = -u$ in $\mathbb{F}_3$, whence $u = 0$ and $x = 0$.

Every column of H_t has sum $7 = 1$ in $\mathbb{F}_3$, so $H_t^{\mathsf{T}}e = j$. The diagonal entries of $H_t^{\mathsf{T}}H_t$ are also 1. Off the diagonal they are 1 at intersection size one and 0 at intersection size three. Consequently

$$A_t = J_n - H_t^{\mathsf{T}}H_t = H_t^{\mathsf{T}}(J_{23-t} - I_{23-t})H_t. \tag{2.1}$$

For $m = 23$ or 21, the matrix $J_m - I_m$ is invertible over $\mathbb{F}_3$. In fact, $(J_m - I_m)x = 0$ implies $x = ue$ and $u = mu$; since $m - 1 \neq 0$ in both cases, $u = 0$. For $m = 22$, its kernel is $\mathbb{F}_3 e$, its image is $e^{\perp}$, and it acts as $-I$ on that image. The image assertion follows from $e^{\mathsf{T}}(J_m - I_m) = 0$ and the rank $m - 1$. Lemma 2.1 proves every image and dimension in the table.

For $t = 0, 2$, the image in the point space is the whole space and its Gram matrix is nonsingular, proving LCD. For $t = 1$, $G = I + J$ restricts to the ordinary dot product on $e^{\perp}$. This restriction is nondegenerate because $e^{\mathsf{T}}e = 22 = 1$: a vector in its radical belongs both to $e^{\perp}$ and to $\mathbb{F}_3 e$, so vanishes. For $t = 0$ the same argument uses $G = 2I$ and $e^{\mathsf{T}}e = 23 = 2$ to give the asserted splitting. Its constant line is nondegenerate since $j^{\mathsf{T}}j = 253 = 1$.

It remains to identify the complementary code when $t = 1, 2$. By (2.1), $B_t = H_t^{\mathsf{T}}H_t - I_n$. Put $M = J_{23-t} - I_{23-t}$ and $G = H_tH_t^{\mathsf{T}}$. Then

$$A_tB_t = H_t^{\mathsf{T}}M(G - I)H_t = 0 :$$

for $t = 2$, $G = I$; for $t = 1$, $G - I = J_{22}$ and $(J_{22} - I_{22})J_{22} = (22-1)J_{22} = 0$. In the latter case $B_t H_t^{\mathsf{T}} e = H_t^{\mathsf{T}}(G - I)e = H_t^{\mathsf{T}} e = j$; in the former case $A_t j = 2j$, since (2.1) and $H_t j = r_t e = e$ give $A_t j = H_t^{\mathsf{T}}(21 - 1)e = 2j$. Thus in both cases at least one graph degree is nonzero in the field. Theorem 1.2 yields $\operatorname{im} B_t = (\operatorname{im} A_t)^{\perp}$. □

Proposition 2.3. *Suppose, in addition, that the length-253 code in Theorem 2.2 has minimum weight 77 and exactly 46 words of that weight. Then its group of coordinate permutations preserving the code is exactly the group induced on the blocks by* $\mathrm{Aut}(\mathcal{D})$.

Proof. For each of the 23 points p, its incidence row h_p is a codeword of weight 77. The 46 words $h_p, -h_p$ are distinct: the incidence rows are independent by the nonsingular Gram matrix, and a nonzero row with entries in $\{0, 1\}$ cannot equal the negative of another such row over $\mathbb{F}_3$. The hypothesis therefore lists all minimum words. A coordinate permutation preserving the code must permute their 23 supports. These supports are exactly the sets of blocks through the respective points. The induced permutation of points and the given permutation of blocks preserve incidence, so form an automorphism of $\mathcal{D}$. Conversely every design automorphism permutes the incidence rows and hence preserves their span, which is the code by Theorem 2.2. This proves equality of the induced groups. □

The minimum-word hypothesis in Proposition 2.3 requires its own exact certificate. The application also identifies the full design group by literal permutations attaining an incidence-refinement bound, using Theorem 6.4. Neither incidence transfer nor recognition alone establishes irreducibility of the augmentation summand.

3 Binary incidence spaces

Let Ω be an n-element set with $n > 0$. Let V_k be the vector space over $\mathbb{F}_2$ with orthonormal basis e_S indexed by the k-subsets of Ω, including $V_0 = \mathbb{F}_2 e_\varnothing$. Put $V_{-1} = V_{n+1} = 0$. Define

$$d_k e_S = \sum_{x \in S} e_{S \setminus \{x\}}, \qquad u_k e_S = \sum_{x \notin S} e_{S \cup \{x\}}.$$

Thus $u_{k-1} = d_k^{\mathsf{T}}$. The space $W_{k-1} = \operatorname{im} u_{k-1} \leq V_k$ is generated by the indicator vectors of all k-sets containing a fixed $(k-1)$-set, exactly the binary incidence code denoted by W_{k-1} in F017. Its dual is $\ker d_k$.

Lemma 3.1. *The following identities hold, with all scalar integers reduced modulo two:*

$$d_{k-1} d_k = 0, \qquad u_{k+1} u_k = 0, \qquad d_{k+1} u_k + u_{k-1} d_k = n I_{V_k}.$$

The complex is exact. For $1 \leq k \leq n$,

$$\operatorname{rank} d_k = \binom{n-1}{k-1}, \qquad \dim \ker d_k = \binom{n-1}{k}.$$

Proof. Each $(k-2)$-subset of a fixed k-set is obtained by two orders of deletion, so $d^2 = 0$. Transposition gives $u^2 = 0$. In $d_{k+1} u_k e_S$, the coefficient of e_S is $n - k$, while in $u_{k-1} d_k e_S$ it is k. Each other possible term replaces one element of S by an element outside S; it occurs once in each expression and cancels. This proves the third identity.

For exactness, fix $v \in \Omega$ and define $h_k e_S = e_{S \cup \{v\}}$ if $v \notin S$, and $h_k e_S = 0$ otherwise. Directly on a basis vector,

$$d_{k+1} h_k + h_{k-1} d_k = I_{V_k}.$$

If $v \notin S$, deleting the newly added v contributes e_S, and all other terms occur twice. If $v \in S$, only deletion and subsequent addition of v survives. Thus if $d_k c = 0$, then $c = d_{k+1} h_k c$,

proving $\ker d_k = \operatorname{im} d_{k+1}$. Now $\operatorname{rank} d_1 = 1$, since every singleton maps to $e_\varnothing$. Exactness and rank–nullity give $\operatorname{rank} d_{k+1} = \binom{n}{k} - \operatorname{rank} d_k$; induction with Pascal's identity proves both dimension formulas. Transposing also gives $\ker u_k = \operatorname{im} u_{k-1} = W_{k-1}$. □

Theorem 3.2. *Let $1 \le k < n$, let A be the binary adjacency matrix of $J(n,k)$, and put $E = u_{k-1}d_k$. Then*

$$E^2 = nE, \qquad A = E + (k \bmod 2)I.$$

If n is odd, E is the orthogonal projector onto W_{k-1}. Consequently

	$\operatorname{im} A$	$\operatorname{im}(A+I)$
k even	W_{k-1}	$W_{k-1}^\perp$
k odd	$W_{k-1}^\perp$	W_{k-1}.

Both codes are LCD. If n is even, $E^2 = 0$ and $\operatorname{im} E \subseteq W_{k-1} \cap W_{k-1}^\perp$. In that case

	$\operatorname{im} A$	$\operatorname{im}(A+I)$
k even	$\operatorname{im} E$	V_k
k odd	V_k	$\operatorname{im} E$.

Proof. The diagonal entry of $E = d_k^{\mathsf{T}} d_k$ is k. For two distinct k-sets, the entry counts common $(k-1)$-subsets, so it is one precisely when their intersection has size $k-1$. This proves the formula for A. Using Lemma 3.1 in degree $k-1$ gives

$$E^2 = u_{k-1}(nI - u_{k-2}d_{k-1})d_k = nE,$$

and the same identities give $Eu_{k-1} = nu_{k-1}$ and $d_kE = nd_k$. If n is odd, the first identity shows that E acts as the identity on W_{k-1}, while its definition shows that its image is contained there. Since E is symmetric and idempotent, its kernel is the orthogonal complement of its image, and $I + E$ projects onto that kernel. The table for odd n follows. If n is even, $\operatorname{im} E \subseteq \operatorname{im} u_{k-1} = W_{k-1}$ and $d_kE = 0$, giving the hull inclusion. Finally $(I+E)^2 = I$, so $I + E$ is invertible, which proves the second table. □

Theorem 3.3. *For $1 \le k < n$, the binary code W_{k-1} has minimum distance $n - k + 1$. Its minimum words are exactly*

$$u_{k-1}e_R = \sum_{S \supset R,\ |S|=k} e_S, \qquad |R| = k-1.$$

The dual code $\ker d_k$ *has minimum distance $k+1$. Its minimum words are exactly*

$$d_{k+1}e_T = \sum_{S \subset T,\ |S|=k} e_S, \qquad |T| = k+1.$$

Proof. Let $0 \ne c \in W_{k-1} = \ker u_k$, and choose K in its support. For each $x \notin K$, the coefficient of $e_{K\cup\{x\}}$ in $u_k c$ is zero. Since K contributes one, at least one other set $K\setminus\{i\}\cup\{x\}$ belongs to the support. These choices are distinct for different x, so $\operatorname{wt}(c) \ge 1 + (n-k)$. The proposed star words attain this bound.

Suppose equality holds. For each $x \notin K$ there is exactly one chosen $i = f(x) \in K$, and these $n-k$ sets together with K comprise the entire support. If there are at least two elements $x, y \notin K$, apply $u_k c = 0$ to the $(k+1)$-set $K \setminus \{i\} \cup \{x,y\}$ for each $i \in K$. The only possible supported facets are $K \setminus \{i\} \cup \{x\}$ and $K \setminus \{i\} \cup \{y\}$: every other facet contains both outside elements. Hence $f(x) = i$ if and only if $f(y) = i$, and f is constant. If there is only one outside element, f is already constant. The support is therefore the star of $K \setminus \{f(x)\}$, as asserted. This also covers $k = 1$.

Now let $0 \neq c \in \ker d_k$ and choose K in its support. For every $i \in K$, the coefficient of $e_{K\setminus\{i\}}$ in $d_k c$ forces a second supported set containing that facet. Such a set has the form $K \setminus \{i\} \cup \{x_i\}$ with $x_i \notin K$. For distinct i these sets are distinct, so $\mathrm{wt}(c) \geq k+1$. The proposed boundary words attain the bound, since $d_k d_{k+1} = 0$.

At equality, those k sets and K are the full support. If $k \geq 2$, consider distinct $i, j \in K$ and the $(k-1)$-set $K\setminus\{i,j\}\cup\{x_i\}$. It lies in the supported set $K\setminus\{i\}\cup\{x_i\}$. The only possible second supported set containing it is $K\setminus\{j\}\cup\{x_j\}$, and containment forces $x_i = x_j$. Thus all x_i are the same element x, and the support is the boundary of $K \cup \{x\}$. If $k = 1$, every word of weight two is already the boundary of a two-element set. This proves exhaustion. □

4 Ternary incidence on triples

Let Ω have n elements, and let V be the vector space over $\mathbb{F}_3$ with coordinates indexed by the triples of Ω. For each pair $e \subset \Omega$, write $v_e \in V$ for the indicator of the triples containing e. Let W_2 be the span of these vectors and let $\mathbf{j}$ be the constant vector. All orthogonal complements below refer to the usual coordinate inner product.

Lemma 4.1. *If $n \geq 5$, then $\dim W_2 = \binom{n}{2} - 1$ and $\mathbf{j} \notin W_2$.*

Proof. Suppose scalars $a_{ij} = a_{ji}$, indexed by distinct elements, satisfy

$$a_{ij} + a_{ik} + a_{jk} = c \qquad \text{for every triple } \{i,j,k\}.$$

Fix distinct k, l and put $b_i = a_{ik} - a_{il}$ for $i \notin \{k,l\}$. Subtracting the equations for $\{i,j,k\}$ and $\{i,j,l\}$ gives $b_i + b_j = 0$. There are at least three available indices. For any three, these equations give $b_i = -b_j = b_h = -b_i$, hence $2b_i = 0$ and $b_i = 0$. Thus $a_{ik} = a_{il}$ for every allowable i, k, l. All coefficients on edges incident with any fixed vertex are equal, and symmetry implies that all a_{ij} have one common value a. Consequently $c = 3a = 0$.

Applied with $c = 0$, this identifies the kernel of the map $(a_e)_e \mapsto \sum_e a_e v_e$ as the one-dimensional space of constant coefficient vectors. Applied with $c = 1$, it shows that $\mathbf{j}$ has no preimage. Both assertions follow. □

Theorem 4.2. *Let $n \geq 7$ and $n \equiv 5 \pmod 9$. Let A be the adjacency matrix of the Kneser graph on the triples of Ω, so two triples are adjacent if and only if they are disjoint. Then*

$$\mathrm{im}(I + A) = W_2 \oplus \langle \mathbf{j} \rangle, \qquad \mathrm{im}\, A = \big(W_2 \oplus \langle \mathbf{j} \rangle\big)^{\perp}.$$

Both spaces are LCD, and their dimensions are respectively $\binom{n}{2}$ and $\binom{n}{3} - \binom{n}{2}$.

Proof. Let M_s be the incidence matrix whose rows are the s-subsets and whose columns are the triples, for $s = 1, 2$. For two triples whose intersection has size t, inclusion and exclusion gives

$$\mathbf{1}_{t=0} = 1 - t + \binom{t}{2} - \binom{t}{3}.$$

Since $\binom{t}{3}$ is the equality indicator for the two triples,

$$R := I + A = J - M_1^{\mathsf{T}} M_1 + M_2^{\mathsf{T}} M_2. \tag{4.1}$$

For each point i, the sum of the vectors v_e over the pairs e containing i is twice the point-incidence vector. Since two is invertible in $\mathbb{F}_3$, the row space of M_1 is contained in W_2. Equation (4.1) therefore gives $\mathrm{im}\, R \subseteq W_2 + \langle \mathbf{j} \rangle$.

For a pair e and a triple T, direct counting gives

$$(Rv_e)_T = \mathbf{1}_{e \subseteq T} + (n-5)\mathbf{1}_{e \cap T = \varnothing}.$$

The second coefficient vanishes in $\mathbb{F}_3$. Thus $Rv_e = v_e$. Also $R\mathbf{j} = (1 + \binom{n-3}{3})\mathbf{j} = \mathbf{j}$: writing $n = 9m + 5$, the binomial coefficient is $3m(9m + 2)(9m + 1)/2$, which is divisible by three. It follows that R is the identity on $W_2 + \langle \mathbf{j} \rangle$, and hence that its image is exactly that space and $R^2 = R$.

The matrix R is symmetric. Lemma 1.1 gives $\ker R = (\operatorname{im} R)^\perp$ and the orthogonal decomposition; both summands are LCD. Since $A = R - I$, its image is $\ker R$. Lemma 4.1 now gives the two dimensions. □

5 Intersection counts for subset and Hamming graphs

Write $C_M = \operatorname{im} M$ for a symmetric matrix M over $\mathbb{F}_p$. Let A be the adjacency matrix of a regular simple graph on N vertices, of degree ν, and put $B = J - I - A$. Following F017, call C_A RLCD when $C_A = C_{A+I}^\perp$, and CLCD when $C_A = C_B^\perp$. Suppose ordered pairs of vertices are partitioned into classes, indexed by j, on which both the adjacency indicator a_j and the number i_j of common neighbours are constant. Include the diagonal class among them.

Theorem 5.1. *Put $P_j = i_j + a_j$ and $\bar{\nu} = N - 1 - \nu$. Then:*

1. *C_A is RLCD if and only if $P_j = 0$ in $\mathbb{F}_p$ for every j.*
2. *$C_A \perp C_B$ if and only if $\nu - P_j = 0$ for every j. In this event both codes are LCD. They are complementary if and only if $(\nu, \bar{\nu}) \neq (0, 0)$ in the field; otherwise $\mathbb{F}_p^N = C_A \perp \oplus C_B \perp \oplus \langle \mathbf{1} \rangle$.*
3. *C_B is RLCD if and only if $\bar{\nu} - \nu + P_j = 0$ for every j.*

Proof. The entries of $A^2 + A$ are P_j. Thus their vanishing is exactly $A(A+I) = 0$. The matrices $-A$ and $I + A$ are then complementary symmetric idempotents. Conversely the asserted duality implies that all their row inner products vanish. Next $AB = AJ - A - A^2 = \nu J - A - A^2$, proving the orthogonality test. The remaining claims in part 2 follow from Theorem 1.2, applied to this actual pair of matrices; no condition on their complex spectra is needed. Finally $B(B + I) = B(J - A) = \bar{\nu} J - AB$, which gives part 3 by the same reflexive argument. □

Uniform subset graphs. Let $n \geq 2k + 1$, $k \geq 2$ and $0 \leq r < k$. The vertices of $\Gamma(n, k, r)$ are the k-subsets of an n-element set; adjacency means intersection of size r. For two vertices meeting in j elements, the common-neighbour count is

$$i_j = \sum_a \binom{j}{a} \binom{k-j}{r-a}^2 \binom{n-2k+j}{k-2r+a}, \qquad 0 \leq j \leq k. \tag{5.1}$$

Here binomial coefficients with invalid lower argument are zero. To prove the formula, partition the ground set into the intersection, the two differences and the outside. A common neighbour chooses a elements from the intersection, $r - a$ from each difference and the remaining $k - 2r + a$ outside. These choices are disjoint and exhaustive. In particular $i_k = \nu = \binom{k}{r}\binom{n-k}{k-r}$ and $P_j = i_j + \mathbf{1}_{j=r}$, so Theorem 5.1 applies directly.

For the Johnson graph $r = k - 1$, the only nonzero counts off the diagonal are $i_{k-2} = 4$ and $i_{k-1} = n - 2$. Thus RLCD holds exactly for $p = 2$ and odd n. Under those conditions $\nu = 0$, so CLCD holds exactly when $\binom{n}{k}$ is even. For $k \geq 3$, complement orthogonality itself forces $\nu = 0$ from $P_0 = 0$; then $P_{k-2} = 4$ and $P_{k-1} = n - 1$ force the same characteristic and parity. For $k = 2$, subtraction of the two off-diagonal tests yields $n = 5$ in the field, while $\nu = 4$ gives $2(n - 2) = 4$, hence characteristic two and odd n. The binary dimensions and minimum words follow from Theorems 3.2 and 3.3, not from this count alone.

The domain qualification matters. F017 Corollary 1(1) does not state it explicitly. For example $J(4,3) = K_4$ is RLCD in characteristic three. More generally the adjacency matrix of $J(n, n-1) = K_n$, for $n \geq 3$, is $J - I$, so $(J - I)J = (n-1)J$: it is RLCD exactly when $p \mid (n-1)$. Its complement is empty, so it is CLCD exactly when $J - I$ is invertible, equivalently $p \nmid (n-1)$. Thus the unrestricted source exclusion is false. Formula (5.1) itself remains valid for $n \geq k$ when only realized intersections $\max(0, 2k-n) \leq j \leq k$ are tested; the table below retains its stated $n \geq 2k+1$ domain.

For a Kneser graph ($r = 0$), formula (5.1) reduces to $i_j = \binom{n-2k+j}{k}$. If $n = 2k+1$, then $i_0 = 0$ and $P_0 = 1$, so the code is never RLCD. For $k \geq 3$ it is not CLCD either: complement orthogonality would force both $\nu = P_0 = 1$ and $\nu = i_1 = 0$. For $k = 2$ there is an exception. The Petersen graph has $(N, \nu) = (10, 3)$, $P_0 = 1$, $P_1 = 1$ and $P_2 = 3$. Complement orthogonality holds exactly in characteristic two, where $\nu = 1$ and $\bar{\nu} = 0$; hence this code is CLCD. F017 Corollary 1(2) states the exclusion for every k. Its CLCD assertion requires this endpoint correction. Its RLCD exclusion is valid for every $k \geq 2$.

k	r	RLCD	CLCD
2	0	$p = 2$, $n \equiv 3 \pmod 4$	$p = 2$, $n \equiv 1 \pmod 4$
3	0	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 5 \pmod 9$; $p \geq 5$, $n \equiv 5 \pmod p$	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 2, 8 \pmod 9$; $p \geq 5$, $n \equiv 5 \pmod p$
4	0	$p = 2$, $n \equiv 7 \pmod 8$	$p = 2$, $n \equiv 3 \pmod 8$
3	1	$p = 2$, $n \equiv 0 \pmod 4$	$p = 2$, $n \equiv 0, 1 \pmod 4$
4	1	$p = 2$, $n \equiv 1 \pmod 4$; $p = 3$, $n \equiv 6 \pmod 9$	$p = 2$, $n \equiv 1 \pmod 8$; $p = 3$, $n \equiv 6 \pmod 9$

Table 1: Complete parameter conditions for the five families specified in F017, with $n \geq 2k+1$. Each entry is an if and only if statement.

Completeness of the arithmetic classification. Here is a finite exact certificate for Table 1, including the exclusion of all other primes. For each fixed (k, r), regard the P_j and $\nu - P_j$ as polynomials in n over $\mathbb{Q}$ using (5.1). For each list $f_0, \ldots, f_k$, the accompanying program records polynomial coefficients b_j and a monic polynomial g such that

$$\sum_j b_j f_j = g, \qquad g \mid f_j \quad \text{for all } j.$$

Both assertions are verified by exact polynomial arithmetic. Let D be the product of a common denominator of all coefficients of all b_j and a common denominator of all coefficients of all f_j. For $p \nmid D$, the simultaneous congruences are equivalent to $g(n) = 0$. In the two $k = 3, r = 0$ lists, $g = n - 5$; in all other lists, $g = 1$. The resulting exceptional primes, including primes subsequently excluded by the finite tests, are

(k, r)	P_j	$\nu - P_j$
$(2, 0)$	2	2
$(3, 0)$	$2, 3$	$2, 3$
$(4, 0)$	$2, 3$	$2, 3, 17$
$(3, 1)$	$2, 3, 5$	$2, 3, 23$
$(4, 1)$	$2, 3, 5$	$2, 3, 5.$

For an exceptional prime choose the least power $p^a > k$. For $0 \leq b \leq k$, the identity $(1+X)^{p^a} = 1 + X^{p^a}$ over $\mathbb{F}_p$ proves $\binom{m+p^a}{b} = \binom{m}{b}$ modulo p for nonnegative m. Consequently all the counts, the degree and $\binom{n}{k}$ have period p^a modulo p in the domain $n \geq 2k+1$. Testing one

admissible lift of every residue modulo p^a is therefore exhaustive, not a bounded experiment. The program records these residues and applies the constant-line exclusion in Theorem 5.1 to obtain CLCD. For $k = 3, r = 0, p \geq 5, n \equiv 5$, the degree is zero but $\binom{n}{3} - 1 \equiv 9 \neq 0$, so no further exception occurs. This proves the table from its exact arithmetic certificate.

Hamming graphs. Let $H_k(d, q)$ have vertex set $\{0, \ldots, q-1\}^d$, with adjacency at Hamming distance k, where $1 \leq k \leq d$ and $q \geq 2$. If two vertices have distance j, their common-neighbour count is

$$i_j = \sum_a \binom{d-j}{a} (q-1)^a \frac{j!}{b!^2 e!} (q-2)^e, \qquad b = j + a - k, \quad e = 2k - j - 2a, \tag{5.2}$$

where terms with negative factorial arguments are omitted. In their $d - j$ equal coordinates, choose the a positions where a common neighbour differs. Among the j different coordinates, let b agree with the first vertex, b with the second, and e with neither. The distance equations give the stated values of b, e, with $2b + e = j$. Multinomial counting proves (5.2). Theorem 5.1 now applies with $N = q^d$, $\nu = i_0 = \binom{d}{k}(q-1)^k$ and $P_j = i_j + \mathbf{1}_{j=k}$.

Corollary 5.2. *For odd q, every binary distance-k Hamming code is RLCD. For the distance-two binary Hamming graph over $\mathbb{F}_3$, RLCD holds whenever $d \equiv 0 \pmod 3$.*

Proof. Let X_i act as the complete-graph adjacency matrix $J_q - I_q$ in coordinate i and as the identity in all others. Over $\mathbb{F}_2$, odd q gives $X_i^2 = X_i$, and these matrices commute. The distance-k adjacency matrix is $A_k = \sum_{|S|=k} \prod_{i \in S} X_i$. Squaring cancels mixed terms in characteristic two, so $A_k^2 = A_k$ and $A_k(A_k + I) = 0$. For $q = 2, k = 2$, the only possible common-neighbour distances are $0, 2, 4$, with counts $\binom{d}{2}$, $2(d-2)$ and 6, respectively. If $3 \mid d$, these are $0, -1, 0$ modulo three. This is precisely the RLCD test. □

For completeness, the Hamming spectrum used to compute the dimensions comes from the same tensor construction. Over $\mathbb{Q}$, $J_q - I_q$ has eigenvalue $q - 1$ on the constant vectors and -1 on their complement. Choosing t nonconstant tensor factors therefore gives

$$\lambda_t = \sum_i (-1)^i (q-1)^{k-i} \binom{t}{i} \binom{d-t}{k-i}, \qquad m_t = \binom{d}{t} (q-1)^t.$$

Lemma 8.3 sums precisely those multiplicities for which $p \nmid \lambda_t$. In particular the ternary $H_2(6, 2)$ and $H_2(9, 2)$ codes have dimensions 42 and 342. This rank step is justified by the proved LCD property, rather than an assumption of diagonalizability after reduction modulo p.

Theorem 5.3. *Over $\mathbb{F}_2$, the distance-k binary Hamming adjacency matrix satisfies $A_k^2 = \binom{d}{k} I$. For positive even d, the code of the ordinary cube $Q_d = H_1(d, 2)$ is self-dual of dimension 2^{d-1}. Its complement adjacency matrix is invertible, so the code is not the dual of the complement code.*

Proof. Index coordinates by $\mathbb{F}_2^d$. Write T_s for translation by s. Then $A_k = \sum_{\mathrm{wt}(s)=k} T_s$. The translations commute, each satisfies $T_s^2 = I$, and cross terms cancel in the square, proving the first assertion. Write M_t for the adjacency matrix of the ordinary cube Q_t. Order cube vertices by their last coordinate. If d is even,

$$M_d = \begin{pmatrix} M_{d-1} & I \\ I & M_{d-1} \end{pmatrix}, \qquad M_{d-1}^2 = I.$$

The kernel consists exactly of $(x, M_{d-1}x)$, so both nullity and rank are 2^{d-1}. Since $M_d^2 = 0$ and M_d is symmetric, its image equals its orthogonal complement. On the other hand, with $B = J + I + M_d$ we have $J^2 = 2^d J = 0$ and $JM_d = M_d J = dJ = 0$, so $B^2 = I$. □

The conclusion of F017 displays the complement-dual identity for Q_6, Q_8, Q_{10}. Direct reading of the formula confirms the complement bar; Theorem 5.3 disproves that identity and supplies the corrected self-duality statement. The two remaining Hamming examples in that conclusion are instances of Corollary 5.2 and are retained.

6 Finite certificates for modules and automorphisms

The following criteria explain what the finite data in the applications must certify. They use ordinary linear algebra and group actions. A list of computed ranks is evidence only after its matrices have been identified with the specified code, action or quotient.

Lemma 6.1. *Let R be a matrix over F such that $RR^{\mathsf{T}} = 0$. Then $Q = R^{\mathsf{T}}R$ is symmetric, $Q^2 = 0$, and $\operatorname{im} Q$ is self-orthogonal. If $Rj = 0$ and $j^{\mathsf{T}}j = 0$, then $\operatorname{im} Q + Fj$ is self-orthogonal too.*

Proof. We have $Q^2 = R^{\mathsf{T}}(RR^{\mathsf{T}})R = 0$, and hence $\langle Qx, Qy\rangle = x^{\mathsf{T}}Q^2y = 0$. Also $Qj = R^{\mathsf{T}}Rj = 0$, so j is orthogonal to $\operatorname{im} Q$. The remaining pairing is the assumed norm of j. □

Theorem 6.2. *Let $d \geq 1$, and let $V = F^d$ carry a representation of a group generated by $g_1, \ldots, g_s$, with matrices $M_1, \ldots, M_s$. If d^2 products of these matrices, including the empty product if needed, are linearly independent in $\operatorname{End}_F(V)$, then V is absolutely irreducible. It suffices instead to exhibit a rank-one matrix $P = vu$ in the generated unital algebra such that the columns Mv and rows uN, as M, N run over products of the generators, each span a space of dimension d.*

Proof. The first hypothesis says that the generated algebra is $\operatorname{End}_F(V)$. Every group-invariant subspace is stable under this algebra. If it contains a nonzero vector, an endomorphism can send that vector to any specified vector of V, so the subspace is V. Linear independence of the d^2 matrices persists under every field extension: a nonzero full-size minor stays nonzero. The same argument therefore proves irreducibility after extension.

For the second test, select products M_i, N_j so that the d columns M_iv and the d rows uN_j are bases. Their d^2 outer products form a basis of the matrix space: changing either chosen basis to the standard basis is invertible. Each outer product is M_iPN_j and belongs to the generated algebra. The first argument applies. □

Theorem 6.3. *Let V be a finite-dimensional FG-module, with an invariant filtration $0 = V_0 < V_1 < \cdots < V_s = V$ whose factors $D_i = V_i/V_{i-1}$ are simple. Every simple submodule of V is the image of an injective homomorphism from one of the D_i. Consequently, if a semisimple submodule N contains the image of every homomorphism $D_i \to V$, then $N = \operatorname{Soc}(V)$. For this containment it suffices to check every map in a vector-space basis of each $\operatorname{Hom}_G(D_i, V)$.*

Proof. Let $S \subseteq V$ be simple, and choose the least i with $S \subseteq V_i$. The invariant subspace $S \cap V_{i-1}$ is either zero or S; minimality excludes the latter. Projection therefore embeds S in the simple module D_i, and is an isomorphism. Its inverse followed by inclusion in V is the required homomorphism. Under the containment hypothesis every simple submodule lies in N, so their sum lies in N. Conversely a semisimple submodule is a sum of simple submodules and lies in that sum. Finally an arbitrary homomorphism is a linear combination of basis maps; its image lies in the sum of their images. □

In row coordinates the homomorphisms are exactly the solutions of $XP_g = D_gX$ for each chosen generator. Solving the whole homogeneous system provides the required basis. Checking a selected map would not give the conclusion. Invariant prefixes are checked directly on each generator; simplicity of their factors can be certified by Theorem 6.2. Thus the criterion does not require a classification of all irreducible representations of G.

Theorem 6.4. *Let $\mathcal{S}$ be a collection of distinct subsets of a finite set Ω. Form its bipartite incidence graph, with the two parts initially coloured differently. Refine a partition by recording, for every vertex, its old cell and its numbers of neighbours in every old cell, until no cell splits. Choose $x_1, \dots, x_t \in \Omega$ successively, separating x_i from its current cell and refining again. If that cell has size b_i before separation, and all vertices of Ω have different colours at the end, then*

$$|\operatorname{Aut}(\Omega, \mathcal{S})| \le \prod_{i=1}^{t} b_i.$$

In particular this bounds the coordinate-permutation group of any code whose complete set of minimum supports is $\mathcal{S}$.

Proof. An incidence automorphism preserves the initial parts. Whenever it preserves every old cell it also preserves the neighbour counts defining the refined cells. After separating $x_1, \dots, x_{i-1}$, the stabilizer of those points therefore preserves every current cell. Its orbit of x_i has size at most b_i. At the last stage this stabilizer fixes every point of Ω and hence every subset in $\mathcal{S}$. It is trivial. Repeated orbit–stabilizer gives the inequality. A coordinate permutation preserving a code preserves word weights and therefore its complete minimum-support collection, proving the last claim. □

To obtain an equality, the application supplies actual permutations preserving the code and verifies that they generate a subgroup of order $\prod b_i$. The upper bound concerns the full group, independently of how those permutations were discovered. For a graph with adjacency matrix A, every graph automorphism preserves $\operatorname{im} A$; thus the additional inclusions $G \le \operatorname{Aut}(\Gamma) \le \operatorname{Aut}(C_A)$ identify the full graph group whenever both outer groups have that order.

7 Weights and dual weights

The averaging theorem below limits a distance search when the code has a transitive permutation group. A complete distribution requires either enumeration of the whole code or another proof. In particular a count of messages of bounded weight is not a whole-code weight distribution.

Theorem 7.1 (MacWilliams identity)**.** *Let $C \le \mathbb{F}_q^n$ have dimension k, and let A_i be its number of words of weight i. Write $A_j^\perp$ for the corresponding numbers in the orthogonal dual. Then*

$$A_j^\perp = q^{-k} \sum_{i=0}^{n} A_i \sum_{s=0}^{j} (-1)^s (q-1)^{j-s} \binom{i}{s} \binom{n-i}{j-s}.$$

Proof. Choose a nontrivial additive character $\chi : \mathbb{F}_q \to \mathbb{C}^\times$. Such a character is obtained from a nonzero $\mathbb{F}_p$-linear functional to $\mathbb{F}_p$ and a primitive pth root of unity, where p is the characteristic. For a nontrivial character on a finite additive group its sum is zero: translation by an element on which it is not one multiplies the sum by that value and also leaves it unchanged. For fixed $x \in \mathbb{F}_q^n$, the map $c \mapsto c \cdot x$ is either zero on C or is onto $\mathbb{F}_q$, with equally sized fibres. It follows that

$$q^{-k} \sum_{c \in C} \chi(c \cdot x) = \mathbf{1}_{x \in C^\perp}.$$

Multiply by $z^{\operatorname{wt}(x)}$ and sum over all x. The finite sums may be interchanged. For a fixed c of weight i, the coordinate factors in the inner sum are $1 + (q-1)z$ at its zero entries and $1 - z$

at its nonzero entries. Hence

$$\sum_j A_j^{\perp} z^j = q^{-k} \sum_i A_i (1-z)^i (1+(q-1)z)^{n-i}.$$

Taking the coefficient of z^j proves the formula. □

This is a classical identity, included here to make its repeated use explicit. For a matrix with k independent rows, the map from its q^k coefficient vectors to its row space is bijective. Enumerating each coefficient vector once therefore gives the complete A_i. The displayed identity then determines the dual distribution. Integrality, nonnegativity and totals provide consistency checks; these checks alone would not prove that a proposed input distribution belongs to the code.

8 Transitive distance searches and rank

Let $C \leq \mathbb{F}_p^n$ be a nonzero linear code of dimension k. An information set is a set I of k coordinates such that the restriction $C \longrightarrow \mathbb{F}_p^I$ is an isomorphism. Gaussian elimination produces a generator matrix G whose columns indexed by I form the identity. For this choice of G, the message of $c = uG$ is exactly $c|_I = u$.

Theorem 8.1. *Suppose that a group T of coordinate permutations preserves C and is transitive on the n coordinates. Fix an information set I. For every $c \in C$ of weight w, some $t \in T$ satisfies*

$$\mathrm{wt}((tc)|_I) \leq \left\lfloor \frac{kw}{n} \right\rfloor.$$

Consequently, for any positive integer D, the following finite data prove that C has minimum distance D:

1. *a codeword of weight D;*
2. *verification that each nonzero message of weight at most $\lfloor k(D-1)/n \rfloor$ gives a codeword of weight at least D.*

Proof. The permutation group T is finite. For each fixed coordinate x and each y, transitivity implies that exactly $|T|/n$ elements send x to y: each nonempty fibre is a coset of a point stabilizer. Count pairs (t, x) with $x \in \mathrm{supp}(c)$ and $tx \in I$. This gives

$$\sum_{t \in T} \mathrm{wt}((tc)|_I) = \frac{|T|kw}{n}.$$

At least one summand is no larger than the average, proving the assertion. If a nonzero codeword had weight $w < D$, its image under the corresponding permutation would have nonzero message of weight at most $\lfloor kw/n \rfloor \leq \lfloor k(D-1)/n \rfloor$. Its codeword weight is still w, contradicting item 2. Item 1 proves the reverse bound. □

Corollary 8.2. *Under the same hypotheses, suppose the minimum distance is D and S is a T-invariant set of words of weight D in C. If every word of weight D obtained from a message of weight at most $\lfloor kD/n \rfloor$ belongs to S, then S is the complete set of minimum words.*

Proof. Apply Theorem 8.1 to an arbitrary minimum word. Its resulting representative is in S, and T-invariance returns the original word to S. The opposite inclusion is part of the hypothesis. □

The number of messages in the distance test, including zero, is

$$\sum_{i=0}^{\lfloor k(D-1)/n \rfloor} \binom{k}{i} (p-1)^i.$$

The theorem concerns this complete finite set. It does not treat a random sample as a certificate, and its restricted weight counts are not the weight distribution of the whole code. Transitivity is essential. For example, the binary code generated by $(1,0,0,0)$ has $n = 4$, $k = 1$, $d = 1$ and information set $\{1\}$. A putative lower-bound test for $D = 2$ enumerates only the zero message and would give a false conclusion if the transitivity hypothesis were removed.

For a proposed invariant set of minimum words, the hypothesis in Corollary 8.2 can also be checked by counting. If its intersection with the restricted message set has the same size as the complete restricted minimum set, inclusion and equality of these finite sizes prove equality of sets. No disjointness or orbit count is assumed without checking it.

Lemma 8.3. *Let A be a symmetric integer matrix with integral eigenvalues $\lambda_1, \ldots, \lambda_s$ of multiplicities $m_1, \ldots, m_s$ over $\mathbb{Q}$. Suppose that the code $\operatorname{im} \bar{A}$ over $\mathbb{F}_p$ is LCD, where a bar denotes reduction modulo p. Then*

$$\operatorname{rank} \bar{A} = \sum_{p \nmid \lambda_i} m_i.$$

Proof. Symmetry gives $(\operatorname{im} \bar{A})^\perp = \ker \bar{A}$. The LCD hypothesis and rank–nullity therefore give the invariant direct sum $\mathbb{F}_p^n = \ker \bar{A} \oplus \operatorname{im} \bar{A}$. The restriction of $\bar{A}$ to its image is injective, hence invertible. Its characteristic polynomial has nonzero constant coefficient. Thus the multiplicity of zero in the characteristic polynomial of $\bar{A}$ is exactly $\dim \ker \bar{A}$. Reduction of the integer characteristic polynomial gives $\prod_i (X - \bar{\lambda}_i)^{m_i}$, whose zero multiplicity is $\sum_{p \mid \lambda_i} m_i$. Subtracting from n proves the formula. □

Required finite arithmetic. The polynomial identities and complete residue lists for Table 1 are certified by `check_association_counts.py`. The proof explains why these finite identities settle every prime and every admissible parameter, rather than inferring an infinite statement from sampled graphs. The applications specify the further code and group certificates they require. No finite distance, irreducibility or full automorphism assertion follows merely from an adjacency spectrum or a congruence test.